BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: **25–26J–089**

Project Title: **Data-Driven Analysis of University Past Papers and Lecture Content for Curriculum Optimization**

	Identification and Analysis
Data Privacy Concerns	The project uses academic materials such as lecture notes, slides, and past MCQ exam papers. During development, no personal student data is used. For real usage, student responses such as quiz answers and performance data are collected. All personal information is anonymized, and informed consent is obtained before data collection. This ensures that student privacy is protected while supporting personalized learning.
Data Security	To keep the data safe, access to the system is limited to authorized users only. All data processing is performed in a controlled and secure environment to prevent unauthorized access or misuse. Proper input validation and error handling mechanisms are applied across backend APIs to ensure system stability and prevent malicious inputs. During development, testing was conducted in a local and controlled setup to minimize exposure of data.
Bias and Fairness	Since the system relies on lecture materials and past exam data, there may be imbalances in topic coverage. Frequently tested topics may receive higher priority. To address this, the system uses syllabus-based content and adaptive learning based on student performance. This ensures fair support for different types of learners and promotes balanced understanding.
Transparency and Accountability	We maintain clear documentation of all processes, including data preprocessing, topic mapping, study plan generation, and evaluation. The system uses explainable methods such as topic frequency, similarity scores, and performance analysis to generate outputs. Visualizations and structured feedback help users understand how decisions are made. The team is responsible for ensuring the system is accurate, fair, and understandable.
Impact on Stakeholders	The system primarily impacts students. For students, it provides personalized study plans, adaptive quizzes, and targeted feedback, enabling more effective exam preparation. It helps identify weak areas and prioritize important topics, improving study efficiency. By supporting different learning styles through structured and interactive features, the system promotes independent learning and better academic performance.